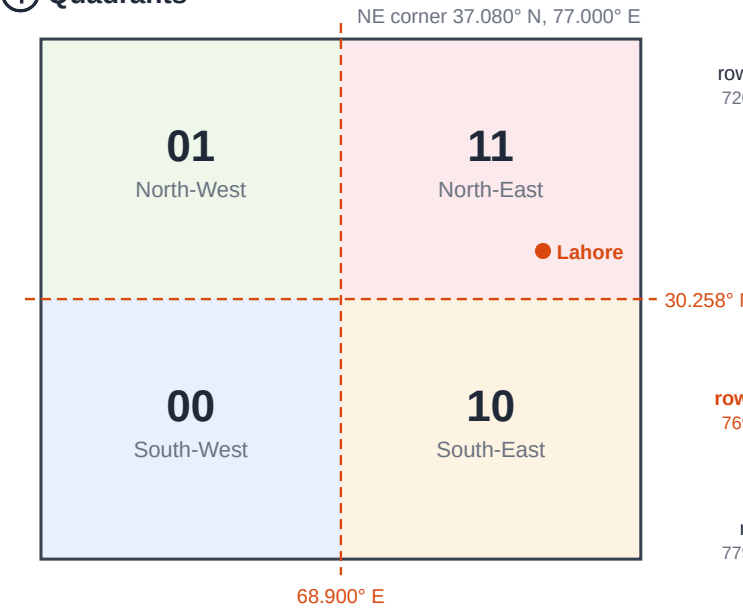
# Pakistan National Grid — how a grid code is built

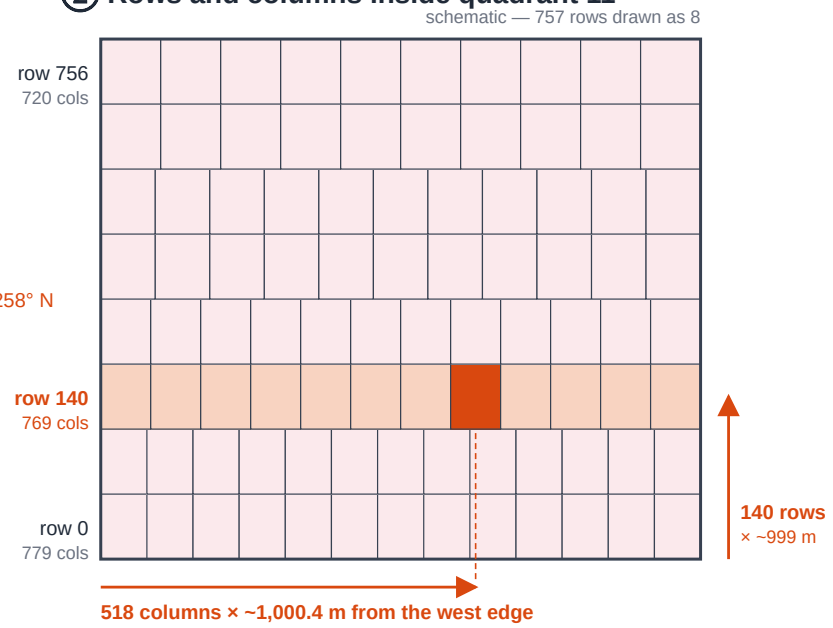
Worked example: central Lahore, 31.5204° N, 74.3587° E → 11-0140-0518-01

## ① Quadrants



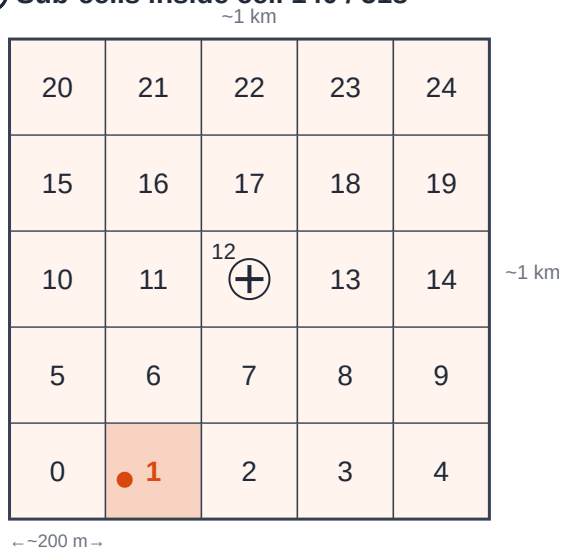

Is it north of 30.258° and east of 68.9°?
**Yes and yes → quadrant 11 (number 3)**

## ② Rows and columns inside quadrant 11



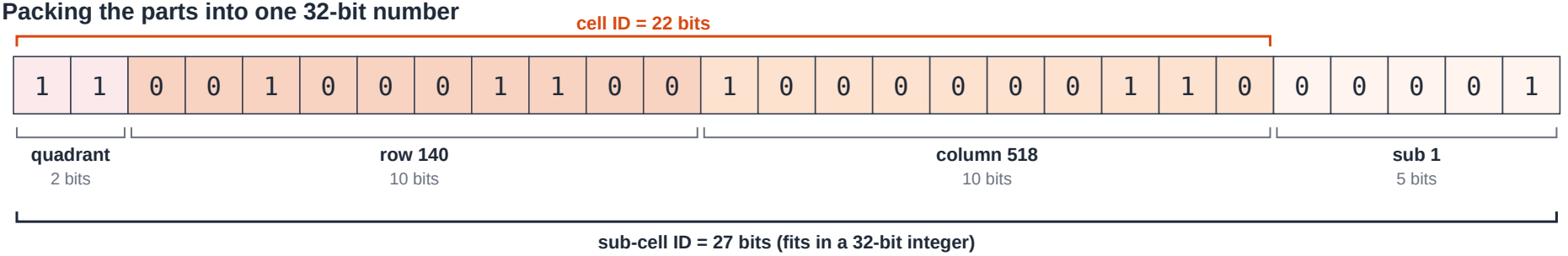

Rows: equal latitude bands, ~1 km tall.
Columns: each row gets its own count so cells stay ~1 km wide; rows further north are narrower, so they get fewer.

## ③ Sub-cells inside cell 140 / 518




● point: 0.08 up, 0.24 across the cell
**sub-row 0, sub-col 1 → sub = 0 × 5 + 1 = 1**
⊕ cell centroid (stored as a GPS point)
Numbering: sub = sub_row × 5 + sub_col (0–24)

## ④ Packing the parts into one 32-bit number




| | |
|---|---|
| **Row** | `floor((31.5204 − 30.258) ÷ 0.0090119) = floor(140.08) = 140` |
| **Column** | `floor((74.3587 − 68.900) ÷ 0.0105332) = floor(518.24) = 518` |
| **Cell** | `(3 « 20) \| (140 « 10) \| 518 = 3,289,606` |
| **Sub** | `(3,289,606 « 5) \| 1 = 105,267,393` |
| **Parent** | `105,267,393 » 5 = 3,289,606 (no lookup needed)` |

Readable code

**11-0140-0518-01**

quadrant · row · column · sub